UNIVERSITY OF COLOMBO SCHOOL OF COMPUTING (UCSC)

SCS2301 / IS2201: Software Engineering Project • Academic Year 2025/2026

Interim Action Plan Document

Project: University Sports Management System (USMS)

GROUP IS03

Proposal Outcome: ACCEPT

Submission Date: Sep 24, 2026

1. EXECUTIVE SUMMARY & EVALUATION CONTEXT

Following the formal Proposal Presentation and Defense for **Group IS03**, the academic evaluation panel accepted the core proposal, formally approving the *Problem Definition and Proposed Solution*, *Project Scope*, *Feasibility Study*, and *Utilization of Development Frameworks*. The overall outcome of the defense was recorded as **ACCEPT**.

The evaluation panel provided constructive, specific feedback regarding practical assumptions surrounding university student database verifications, the formalization of security as explicit functional requirements, adherence to full multi-sport coverage, elimination of automated medical lock releases, and the standardization of UML Design Diagrams (which were flagged for rectification).

This **Action Plan Document** establishes our formal engineering response, technical actions already implemented in the active codebase, and our development roadmap leading to the final defense.

2. PROPOSAL FEEDBACK ANALYSIS & ACTION MATRIX

Evaluation Area	Panel Feedback & Directives	Status	Action Taken & Refinement
Student DB Verification	"You can't cross-check with the student database practically. So, it is okay to assume that the verification against real university students can be done when such information is available."	ADDRESSED	Implemented internal Verification Queue and decoupled API adapter interface.
Security Classification	"Security is a functional requirement."	ADDRESSED	Specified and implemented 6 distinct Functional Requirements (RBAC, BCRYPT, PDO).
Scope: Sports Coverage	"You mentioned that you will be covering all the sports."	ADDRESSED	Normalized relational model & dynamic catalog supporting all 24+ UOC sports.
Medical Lock Release	"You don't have to automate medical lock release as it may result in unnecessary complications."	ADDRESSED	Automated release eliminated; clearance is strictly certified by Medical Officer.
Design Diagrams REJECT	Component diagrams and architectural representations showed discrepancies with component reuse and GitHub UI structure.	RECTIFIED	Standardized UML Component Diagram & unified cross-portal UI design tokens.

3. HIGH-LEVEL SUMMARY OF ACTION PLAN PROGRESS

Implementation Highlights: Over the interim development cycle, Group IS03 has synchronized all team contributions into the `main` branch on GitHub (`Hemsara04/USMS`). All five feedback directives have been systematically addressed: student verification is modeled through an administrative approval workflow, security is implemented as robust functional requirements, medical clearance is certified manually by clinicians, multi-sport scalability is operational, and architecture diagrams have been standardized.

3. DETAILED TECHNICAL RESOLUTION OF PANEL DIRECTIVES

3.1 Practical Student Verification & Institutional Assumption Modeling

> **Panel Understanding:** Undergraduate project teams do not have access to live University of Colombo Student Information System (SIS/ERP) APIs due to institutional security protocols and privacy policies. The panel confirmed it is appropriate to model external verification as an engineering assumption.

Technical Resolution: USMS adopts a resilient **Two-Tier Verification Architecture**:

- **Operational Tier (Internal Verification Queue):** In `views/admin_user_management.php` and `sport_application`, student-athletes provide their institutional Index Number, Academic Year, and Faculty during registration. Records enter a pending administrative queue where Admins and TICs verify submitted credentials and approve/reject applications with full audit logging.
- **Architectural Tier (External Verification Adapter):** The system defines a service adapter interface (`StudentVerificationInterface`). When institutional API access is officially provisioned, the adapter can connect directly without refactoring existing database schemas or controller logic.

3.2 Elevating Security to Explicit Functional Requirements (FR)

Security has been refactored from general non-functional claims into **six demonstrable Functional Requirements**, fully implemented in the active codebase:

FR ID	Functional Requirement	Technical Implementation	Verification Reference
FR-SEC-01	Cryptographic Password & Secret Hashing	BCRYPT adaptive hashing with automatic unique salt generation via PHP `password_hash(..., PASSWORD_BCRYPT)`.	`models/User.php:68, 93, 136` `models/Admin.php:198`
FR-SEC-02	Role-Based Access Control (RBAC)	Strict authorization enforcement across 8 system roles (Admin, Coach, TIC, Medical Officer, President, Athlete, etc.).	`config/auth_guard.php` executes `requireRole()` on Line 1 of all views.
FR-SEC-03	Session Lifecycle & Invalidation	Secure session generation upon login, session hijacking defense, and complete cookie/session destruction on logout.	`AuthController.php:42-97` and `AuthController.php:327-340`
FR-SEC-04	SQL Injection (SQLi) Immunity	100% Parameterized Prepared Statements across all queries; elimination of raw query string concatenation.	`config/db.php` enforces `PDO::ATTR_EMULATE_PREPARES => false`.
FR-SEC-05	Cross-Site Scripting (XSS) Sanitization	Strict context-aware character escaping for all database and user inputs before rendering into the DOM.	Universal application of `htmlspecialchars(..., ENT_QUOTES)` in views.
FR-SEC-06	Append-Only Administrative Audit Trail	Logging of security-critical administrative activities (role modifications, account deletions, verifications).	Maintained in `views/admin_audit_logs.php` logging user, timestamp, IP, and status.

3.3 Multi-Sport Architecture & Complete Scope Coverage

The panel reinforced that USMS must comprehensively support **all recognized university sports**. Our database schema and software architecture achieve complete multi-sport scalability through normalized entities:

- **Dynamic Sports Entity (`sports`):** Maintains program categories (Indoor, Outdoor, Racquet, Aquatic), program types (Men's Varsity, Women's Varsity, Co-ed Club), and roster constraints.
- **Relational Decoupling:** Operational entities—`training_sessions`, `fixtures`, `match_squads`, `equipment`, and `injuries`—are relationally mapped to `sport_id` via indexed foreign keys rather than static enums.
- **Administrative Sports Catalog:** Via `views/admin_sports_catalog.php`, athletic administrators can register and configure new university sports dynamically without source code modification.

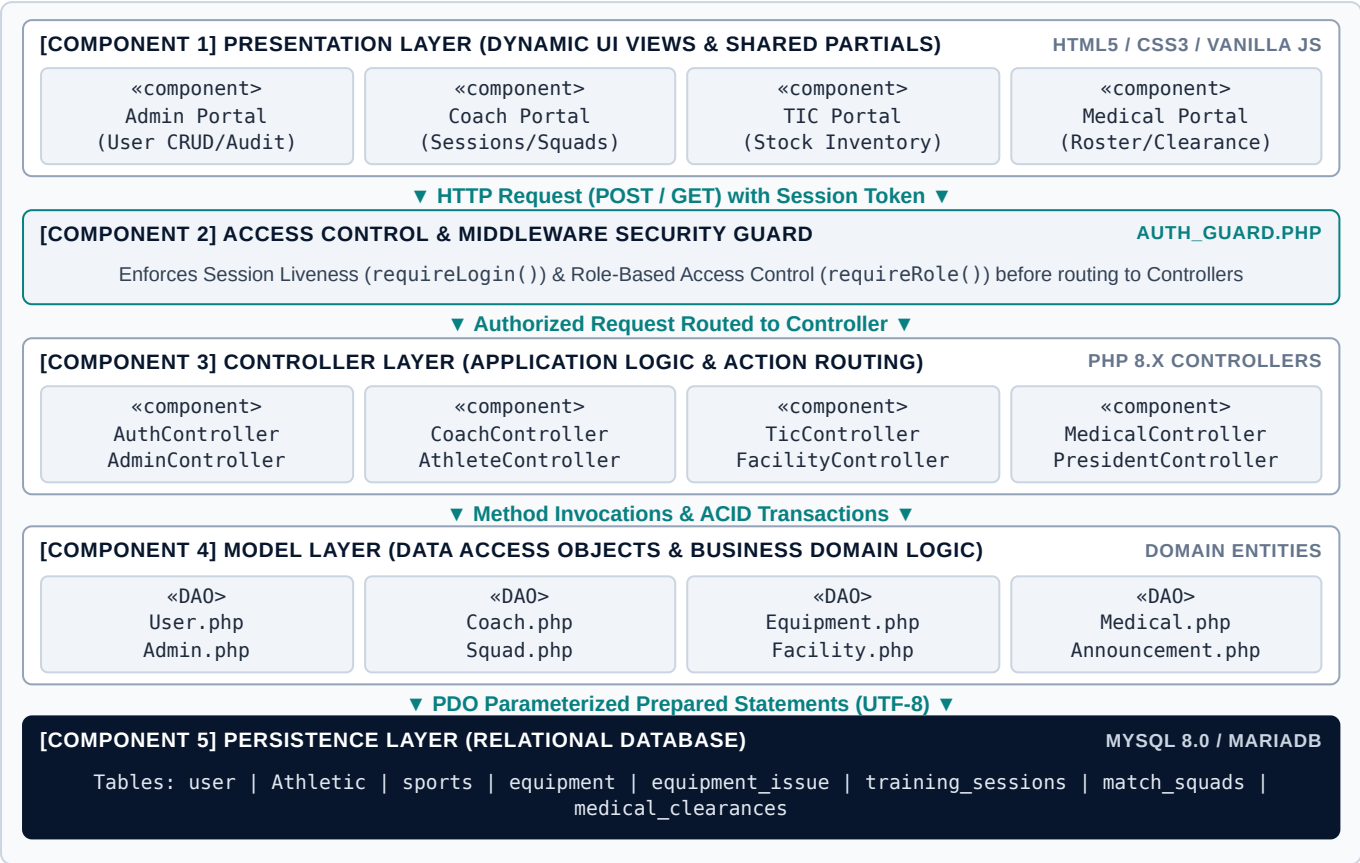
3.4 Clinical Discretion vs. Automated Medical Lock Release

> **Hazard Identified by Panel:** Automating medical clearance based solely on elapsed recovery time introduces severe clinical and institutional liability. An athlete could return to competition with unresolved physical trauma.

Remediation Implemented: Automated lock expiration has been **permanently removed**. Medical clearance is strictly restricted to certified clinicians through the **Medical Officer portal** (`views/med_clearance.php` and `models/Medical.php`).

3.5 Rectifying Design Diagrams: Standardized UML Component Architecture

The panel's rejection of Design Diagrams was rooted in architectural divergence between early diagrams and component reusability. We have restructured our system architecture into a formal **UML Component Architecture** strictly reflecting our active Model-View-Controller (MVC) implementation:



UI Component Harmonization Across GitHub: All interfaces across team feature branches adhere to a synchronized Design System: Deep Navy `#07162d` and Athletic Teal `#087f80` tokens, with reusable partials (e.g., `admin_sidebar.php`, `tic_sidebar.php`) ensuring identical structure across all portals.

4. CORE CRUD MODULE TRACEABILITY (IMPLEMENTED MODULES)

User Role	Primary CRUD Module	Files & Implementation	Key Features & Safeguards

System Admin	User Management CRUD & Account Provisioning	`views/admin_user_management.php`
		`controllers/AdminController.php`
		`models/Admin.php`
			Account creation, role assignment, password resets, self-deletion prevention, verification queue.

TIC Officer	Equipment & Stock Inventory CRUD	`views/tic_stock_inventory.php`
		`controllers/TicController.php`
		`models/Equipment.php`
			Smart restock auto-detection, condition tracking, ACID transactional checkouts and returns.

Coach	Training Sessions & Squad Selection CRUD	`views/coach_training_sessions.php`
		`controllers/CoachController.php`
		`models/Coach.php`
			Practice scheduling, venue booking, attendance sheet generation, match squad formulation.

Medical Officer	Athlete Health Registry & Injury Tracking CRUD	`views/med_athlete_roster.php`
		`controllers/MedicalController.php`
		`models/Medical.php`
			Clinical health profile updates, active injury logging, manual medical clearance certification.

5. WORK BREAKDOWN STRUCTURE (WBS) & IMPLEMENTATION MILESTONES

To ensure structured execution between the interim review and final submission, the development trajectory is divided into prioritized, measurable sprints:

Milestone	Planned Activities & Engineering Tasks	Deliverables / Artifacts	Status
Sprint 1	Database schema normalization, Unified Authentication Portal, BCRYPT hashing, RBAC middleware.	`config/auth_guard.php`, `views/login.php`	COMPLETED
Sprint 2	Admin Portal Full User CRUD (Provision, Edit, Delete), Verification Queue, Metrics Dashboard.	`views/admin_user_management.php`, `models/Admin.php`	COMPLETED
Sprint 3	TIC Stock Inventory Smart Restock CRUD, Equipment Checkout transactions, Medical Manual Clearance.	`views/tic_stock_inventory.php`, `models/Equipment.php`	VERIFIED
Sprint 4	Comprehensive UML diagram overhaul (Component, Activity, and Sequence diagrams aligned with MVC).	Revised Architecture & Design Document	IN PROGRESS
Sprint 5	Cross-portal UI component alignment, responsive mobile optimization, end-to-end user testing.	Quality Assurance & Usability Report	SCHEDULED
Final Phase	Deployment verification, final technical documentation, video demonstration, and system defense.	Production Release & Final Defense	SCHEDULED

6. GITHUB REPOSITORY STRATEGY & VERSION CONTROL INTEGRITY

Our team enforces strict version control practices on GitHub (**Hemsara04/USMS**). Features are developed in isolated contributor branches (`hasitha`, `sandaruwan`, `abdullah`) and integrated into `main` strictly via reviewed Pull Requests. All security safeguards, full CRUD modules, and design revisions cited in this document are committed, syntax-checked, and operational on the `main` branch.

7. CONCLUSION & COMMITMENT

Group IS03 has systematically transformed all evaluator feedback into actionable engineering implementations. By establishing pragmatic student verification assumptions, elevating security to demonstrable functional requirements, committing to full multi-sport architecture, enforcing manual clinical clearance, and standardizing our UML design diagrams, USMS is on schedule for high-caliber final delivery.

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